

Statistical Modeling

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1 General Methods in Statistical Modeling

Given a collection of pairs of data $\{(x_i, y_i) \in \mathbb{R}^n \times \mathbb{R} \mid i \in I\}$, we would like to ask whether there is a relation between x_i and y_i . Formally, we may ask for a simple function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $y_i = f(x_i)$ for all $i \in I$. We may then ask how close does the approximation f models the actual relation between x_i and y_i .

There are three components to the problem. One is to ask for candidates of the function f that describes the relation, the second is to find the best fitting parameters for the function f , the third is to measure how accurate the best fitting parameters estimate the relation.

1.1 Estimators

Definition 1.1.1 (Least Square Estimation)

Definition 1.1.2 (Maximum Likelihood Estimation)

1.2 Hypothesis Testing

Definition 1.2.1 (Null Hypothesis)

Definition 1.2.2 (Alternative Hypothesis)

2 Regression Analysis

We use regression analysis in a certain pattern recognition problem when the dependent variable is real-valued. It is the first step of the pipeline, i.e. regression models propose their respective functions f that models the relation.

Regression models often involve the following components.

- The unknown parameters β .
- The independent variables X_i .
- The dependent variables Y_i .
- The error term e_i .

Most regression models proposes the existence of a function f such that

$$Y_i = f(X_i, \beta) + e_i$$

for all i .

To employ regression analysis, researchers often assume the following are true.

- The sample is representative of the population at large.
- The independent variables are measured without error.
- Derivations from the model have an expected value of 0, conditional on the covariates.
- Homoscedasticity (the variance of all random variables are finite and equal)
- The error terms are uncorrelated to each other.

2.1 Linear Regression

Definition 2.1.1 (Linear Regression Model) A linear regression model assumes the relationship between the dependent random variable Y and the independent random variables $X_1, \dots, X_n$ are linear. In other words, there exists $\beta_0, \dots, \beta_n \in \mathbb{R}$ together with an unobserved random variable ε such that

$$Y = \beta_0 + \sum_{k=1}^n \beta_k X_k + \varepsilon$$

2.2 Logistic Regression

3 Statistical Classification

We use statistical classification methods in a certain pattern recognition problem when the dependent variable is discrete.

3.1 Classification following Logistic Regression